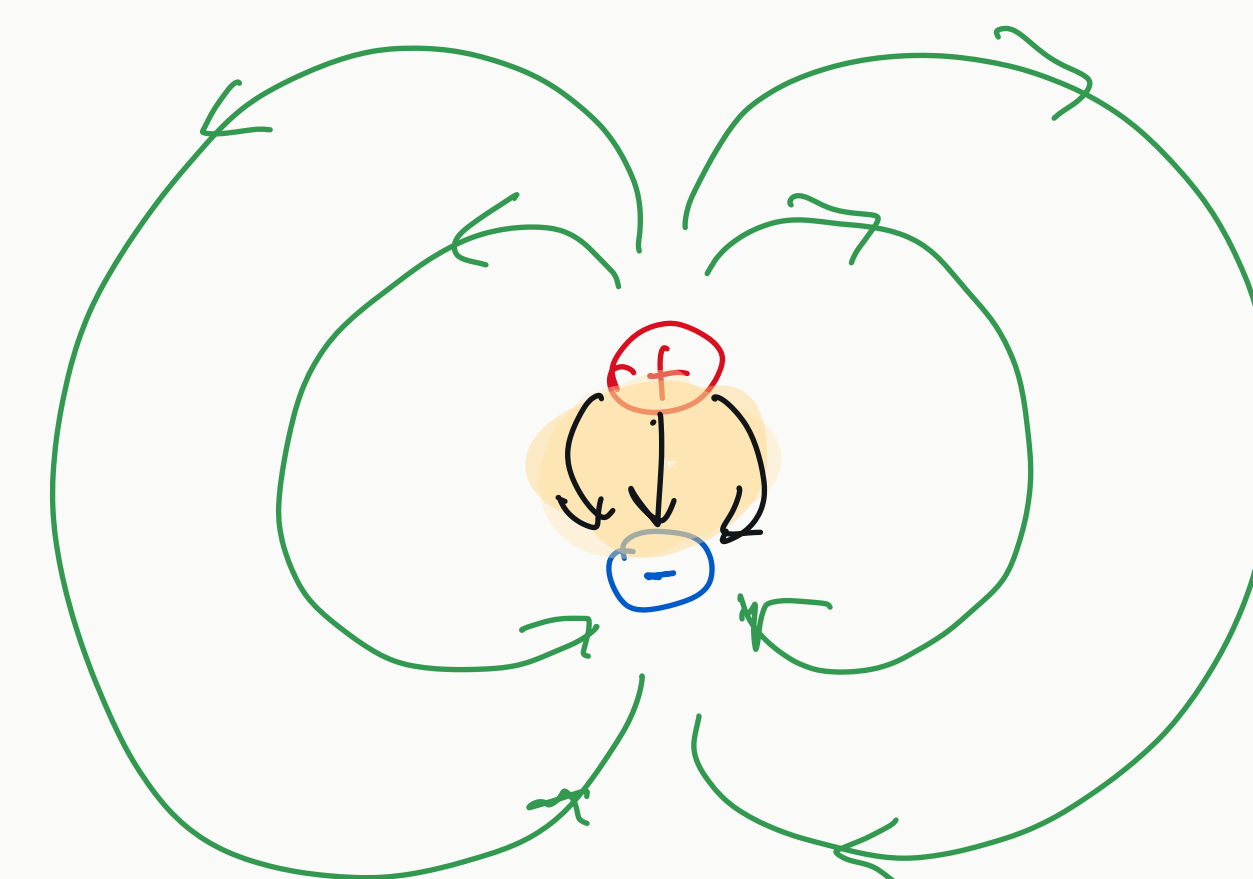


Dipolfeld (fortf.)

Die vollständige Formel für das Dipolfeld:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\underbrace{\frac{3(\vec{r} \cdot \vec{p})\vec{r} - r^2\vec{p}}{r^5}}_{\text{Herleitung}} - \underbrace{\frac{4\pi}{3}\vec{p} \cdot \delta(\vec{r})}_{\text{Herleitung}} \right] \quad (*)$$



Herleitung

$$\int d^3r \vec{E}_{\text{fern}}(\vec{r}) = \vec{0}$$

✓
isotrop

↑
ohne
Beweis

Aber

$$\int \vec{E}(\vec{r}) d^3r \neq \vec{0}$$

✓

✓



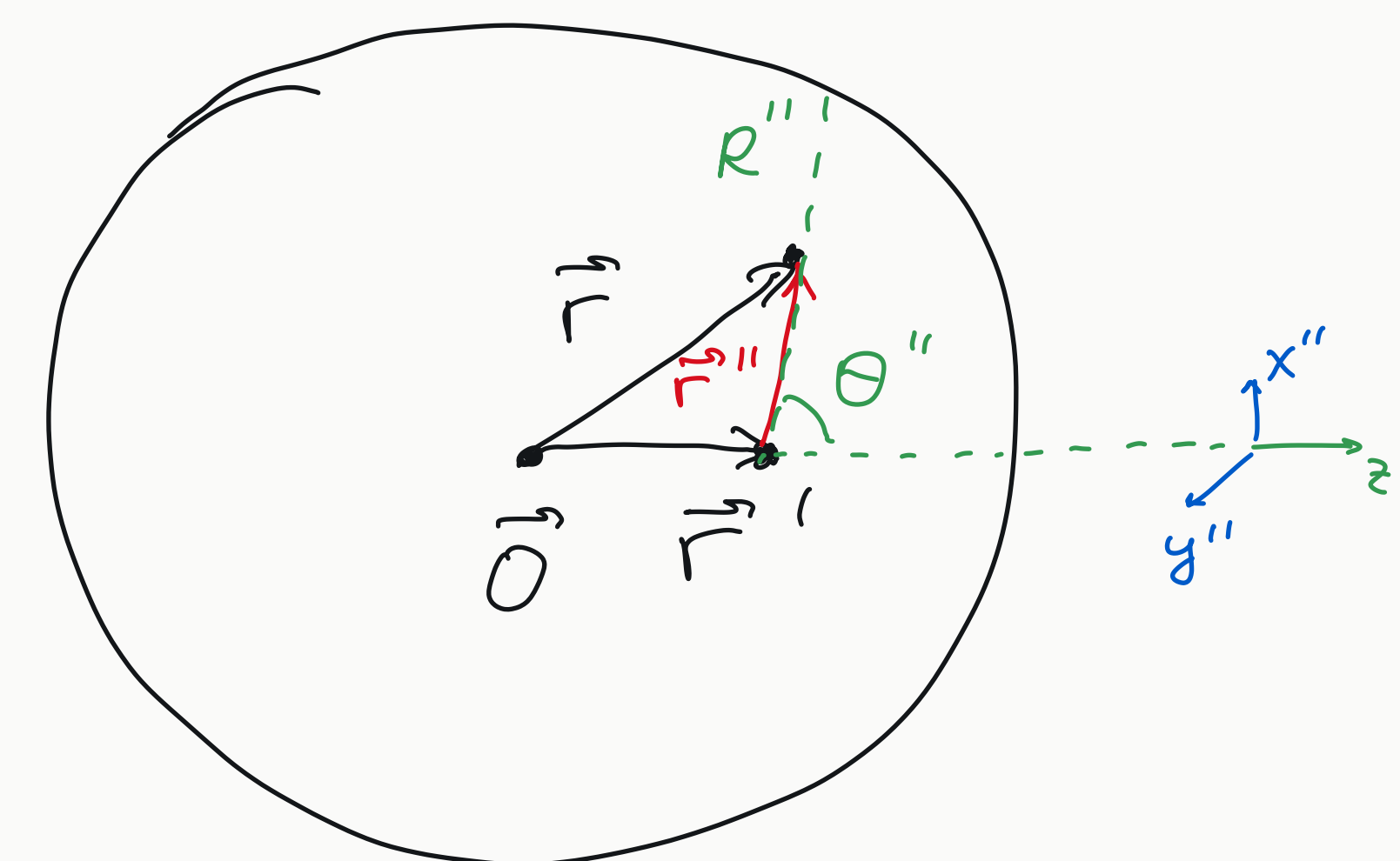
V: Kugel mit dem Radius R

$$\int_V \vec{E}(\vec{r}) d^3r = \int_V d^3r \int_V d^3r' \frac{1}{4\pi\epsilon_0} \frac{\rho(\vec{r}') (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} = \frac{1}{4\pi\epsilon_0} \int_V d^3r' \rho(\vec{r}') \underbrace{\int_V d^3r \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}}_{\vec{I}_1}$$

$$\vec{r}'' = \vec{r} - \vec{r}' \quad d^3r \rightarrow d^3r''$$

$$\vec{r}'' = \frac{\vec{r}}{r} = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|}$$

$$\vec{I}_1 = \int_V d^3r'' \frac{\vec{n}''}{r''^2} = \int_V \underbrace{\cancel{r''^2} dr'' d\Omega''}_{r''^2 \sin\theta'' d\theta'' d\varphi'' dr''} \frac{\vec{n}''}{\cancel{r''^2}} = \int_{\Omega} d\Omega'' \vec{n}'' \underbrace{\int_0^{R''(\vec{n}'')} dr''}_{R''(\vec{n}'')}$$



Axiale Symmetrie: $I_{1x''} = I_{1y''} = 0$

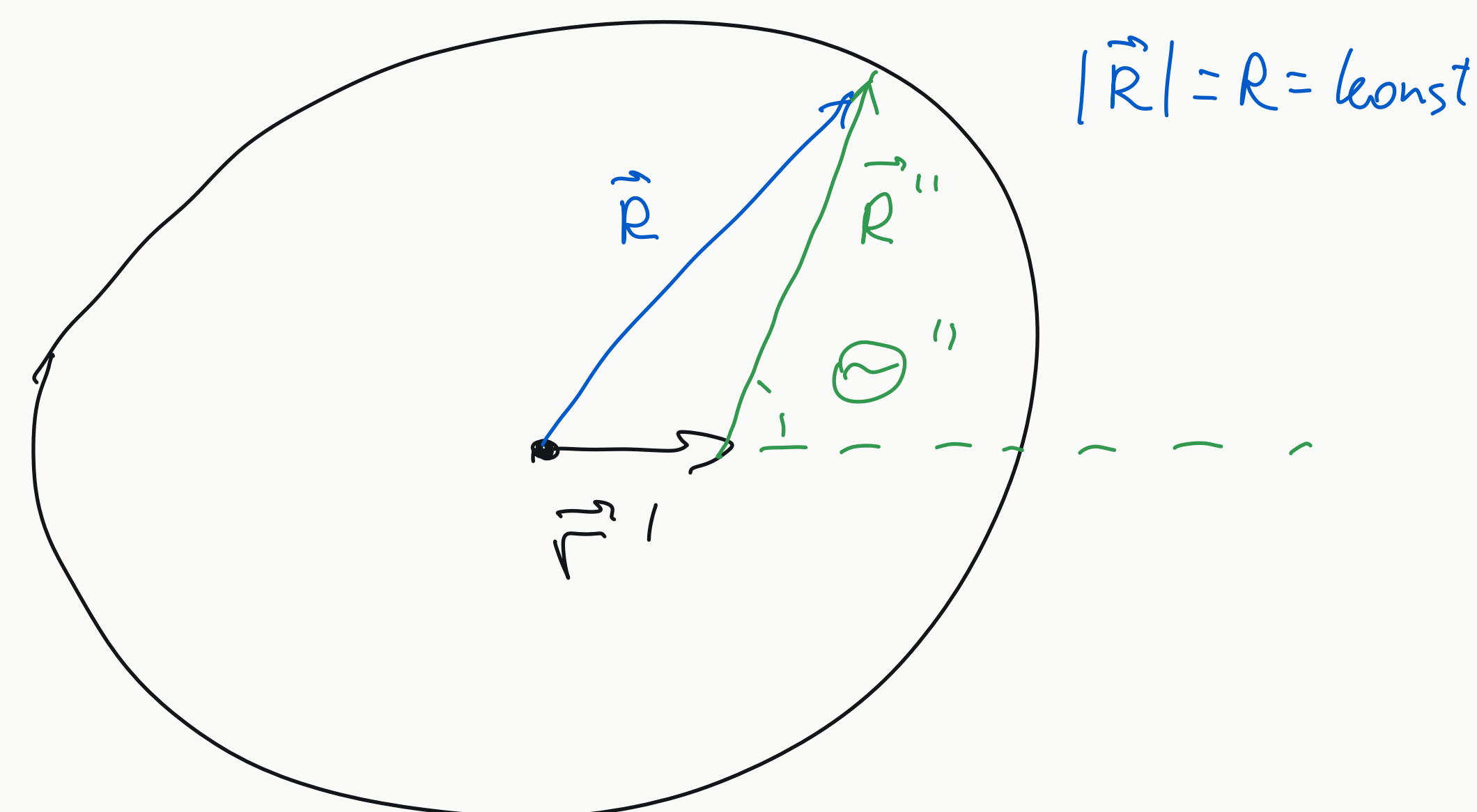
$$I_{1,z''} = \int_{\Omega} d\Omega'' \underbrace{|\vec{r}''|}_{1} \cos \Theta'' \underbrace{R''(\Theta'')}_{2\pi} = \int d\varphi'' \sin \Theta'' d\Theta'' \cos \Theta'' R''(\Theta'')$$

Cosinussatz:

$$R^2 = R''^2 + r'^2 + 2R''r' \cos \Theta'' \Leftrightarrow (\vec{R})^2 = (\vec{r}' + \vec{R}'')^2$$

$$R''^2 + 2r' \cos \Theta'' R'' + r'^2 - R^2 = 0$$

$$R'' = -r' \cos \Theta'' \pm \sqrt{r'^2 \cos^2 \Theta'' + (R^2 - r'^2)}$$



$$I_{1,z''} = 2\pi \int_0^\pi d\Theta'' \sin \Theta'' \cos \Theta'' \left(-r' \cos \Theta'' + \sqrt{r'^2 \cos^2 \Theta'' + R^2 - r'^2} \right) =$$

$$u = \cos \Theta''$$

$$du = -\sin \Theta'' d\Theta''$$

$$\int_0^\pi \rightarrow \int_1^{-1} \rightarrow - \int_{-1}^1$$

$$= 2\pi \int_{-1}^1 du u \left(-r' u + \sqrt{r'^2 u^2 + R^2 - r'^2} \right) =$$

$$\int_V \vec{E}(\vec{r}) d^3r = \frac{1}{4\pi\epsilon_0} \left(-\frac{4\pi}{3} \right) \int d^3r' \rho(\vec{r}') \vec{r}' = \left[\begin{aligned} &= -2\pi r' \int_{-1}^1 du u^2 + 2\pi \int_{-1}^1 du u \sqrt{r'^2 u^2 + R^2 - r'^2} = -\frac{4\pi}{3} r' \\ &\quad \underbrace{\frac{u^3}{3} \Big|_{-1}^1 = \frac{2}{3}}_{\text{antisymmetrisch}} \end{aligned} \right]$$

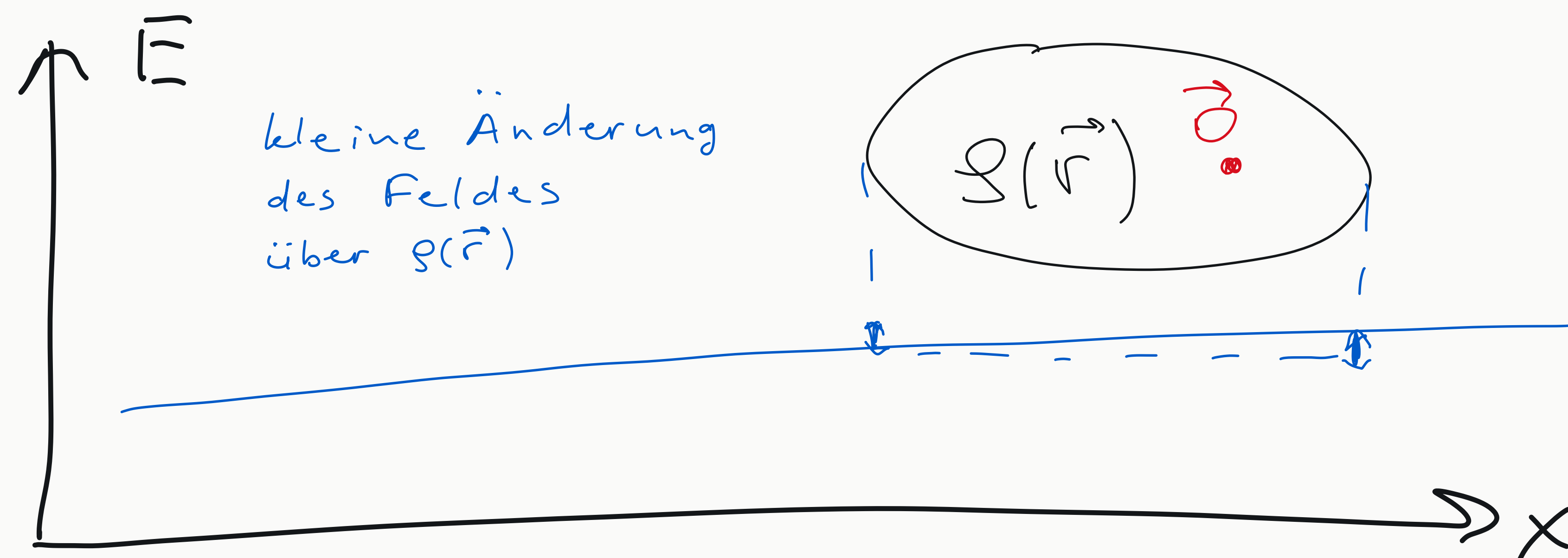
Dipolmoment $\vec{p}$

$$= -\frac{1}{4\pi\epsilon_0} \frac{4\pi}{3} \vec{p} \rightarrow \text{Gl. (*)}$$

$\Downarrow$
 $\int = 0$

$$\vec{I}_1 = -\frac{4\pi}{3} \vec{r}'$$

Ladungsverteilung im äußeren Feld



Pot. Energie

$$W = \int \rho(\vec{r}) \Phi(\vec{r}) d^3r$$

$$\Phi(\vec{r}) = \Phi(\vec{0}) + \underbrace{\frac{\partial \Phi}{\partial r_i}}_{-\vec{E}(\vec{0})} \cdot \vec{r} + \frac{1}{2} \underbrace{\frac{\partial^2 \Phi}{\partial r_i \partial r_j}}_{-\frac{\partial E_i}{\partial r_j} \text{ "Gradient des el. Feldes"}} \bigg|_{\vec{r}=\vec{0}} r_i r_j$$

$$W = \underbrace{\Phi(\vec{0}) \int \rho(\vec{r}) d^3r}_Q - \underbrace{\vec{E}(\vec{0}) \cdot \int \rho(\vec{r}) \vec{r} d^3r}_{\vec{p}} - \underbrace{\frac{1}{2} \frac{\partial E_i}{\partial r_j} \bigg|_{\vec{r}=\vec{0}} \int \rho(\vec{r}) r_i r_j d^3r}_{\frac{1}{2} \frac{\partial E_i}{\partial r_j} \bigg|_{\vec{r}=\vec{0}} \frac{1}{3} \int \rho(\vec{r}) 3 r_i r_j d^3r - 0}$$

$$\textcircled{A} + \textcircled{B} =$$

$$= \frac{1}{6} \frac{\partial E_i}{\partial r_j} \bigg|_{\vec{r}=\vec{0}} \underbrace{\int \rho(\vec{r}) [3 r_i r_j - r^2 \delta_{ij}] d^3r}_{Q_{ij}}$$

$$\frac{\partial E_i}{\partial r_i} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \text{div } \vec{E} = 0$$

$$-\frac{1}{2} \frac{\partial E_i}{\partial r_j} \bigg|_{\vec{r}=\vec{0}} \cdot \frac{1}{3} \int \rho(\vec{r}) \delta_{ij} r^2 d^3r$$

$\textcircled{B}$

$\textcircled{A}$

$$W = q \Phi(0) - \vec{p} \cdot \vec{E}(\vec{0}) - \frac{1}{6} Q_{\alpha\beta} \left. \frac{\partial E_{\alpha}}{\partial r_{\beta}} \right|_{\vec{r}=\vec{0}} + \dots$$

Wechselwirkung zwischen zwei Dipolen

$$W = q \Phi(0) - \underbrace{\vec{p} \cdot \vec{E}(\vec{0})} - \frac{1}{6} Q_{\alpha\beta} \frac{\partial E_{\alpha}}{\partial r_{\beta}} \Big|_{\vec{r}=\vec{0}}$$

$\vec{E}(\vec{0}) \leftarrow$ von anderem Dipol $\vec{p}'$

$$\vec{r}' \neq \vec{r}$$

$\Downarrow$

$$\Delta = \delta(\vec{r} - \vec{r}') = 0$$

$$\vec{E}(\vec{0}) = \frac{1}{4\pi\epsilon_0} \frac{3(\vec{R} \cdot \vec{p}') \vec{R} - R^2 \vec{p}'}{R^5}$$

$$W_{\text{dip}} = \frac{1}{4\pi\epsilon_0} \frac{R^2 \vec{p} \cdot \vec{p}' - 3(\vec{p} \cdot \vec{R})(\vec{p}' \cdot \vec{R})}{R^5}$$

Zum Weiterlesen:

Jackson 4.1, 4.2